

Question ID: 699af7b3

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	Hard

Question

The weight of sample A is **468%** of the weight of sample B, and the weight of sample A is **0.780%** of the weight of sample C. If the weight of sample C is ***p*%** of the weight of sample B, what is the value of ***p***?

Correct Answer: 60000

Rationale

The correct answer is **60,000**. It's given that the weight of sample A is **468%** of the weight of sample B and **0.780%** of the weight of sample C. Let the weight of sample B be ***b*** and the weight of sample C be ***c***. Then, the weight of sample A is equivalent to $\frac{468b}{100}$, or **4.68*b***, and also equivalent to $\frac{0.780c}{100}$, or **0.00780*c***. Setting these expressions equal to each other yields **4.68*b* = 0.00780*c***. Dividing both sides of this equation by **0.00780** yields $\frac{4.68b}{0.00780} = c$, or ***c* = 600*b***. If the weight of sample C is ***p*%** of the weight of sample B, then ***c* = $\frac{p}{100}b$** . Since ***c* = 600*b*** and ***c* = $\frac{p}{100}b$** , it follows that **600*b* = $\frac{p}{100}b$** , or **600 = $\frac{p}{100}$** . Multiplying both sides of this equation by **100** yields **60,000 = *p***. Therefore, the value of ***p*** is **60,000**.